GE LITE

Living in the IT Era

Reviewer for Modules 1 to 3

Module 1: Introduction to Information and Communication Technology
Module 2: History of Computer, Basic Computing Periods
Module 3: The Web and the Internet

Inside: 13 topic summaries, timelines, key facts, kid-simple explanations, and a 40 item practice exam with answer key.

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Exam tips

Know the 'firsts' in Topic 7, the four ages in Topic 8, the five generations in Topic 9, and the difference between Web 1.0, 2.0 and 3.0 in Topic 10. Highlighted words like **this** are the key terms to memorize.

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MODULE 1: INTRODUCTION TO ICT

Information and Communication Technology (ICT)

- **ICT** stands for Information and Communications Technology. It is the infrastructure and components that enable modern computing.
- ICT means all **devices**, **networking components**, **applications** and **systems** that together let people and organizations (businesses, nonprofits, governments) interact in the digital world.
- There is no single, universal definition of ICT.
- Long ago people communicated by letters sent through post mail. Today ICT lets us reach loved ones far away using cellular phones, chat, e-mail, voice mail and social networking.
- ICT is the cheapest means of communication.
- Module objectives: be ready for the semester, know your learning outcomes, and know the role and importance of ICT.

KEY FACTS

ICT Information and Communications Technology	**It includes** Devices, networks, applications and systems
Past Letters sent through post mail	**Present** Phones, chat, e-mail, voice mail, social media

Explain like I'm 5: ICT is all the gadgets and tools that help people share information, like phones, computers and the internet. Long ago people mailed letters and waited for days. Now you can talk to someone far away in one second.

2

MODULE 1: INTRODUCTION TO ICT

Uses of ICT in Our Daily Lives

- **Communication**: chatting, e-mail, voice mail and social networking keep people in touch. ICT also lets students manage their own learning, think critically, work together and see issues with a global perspective.
- ICT gives students in remote areas access to expert teachers and learning resources.
- **Job opportunities**: ICT helps organizations run efficiently, so staff with ICT skills are vital and employers now expect basic ICT skills.
- Example: many pharmacies use robot technology to pick prescribed drugs, so highly trained staff can focus on dispensing and checking medication.
- You can now look for a job from home using a smartphone, laptop or desktop.
- **Education**: ICT helps learning when teachers are digitally literate. Examples: interactive digital whiteboards, students using their own smartphones in class, and the **flipped classroom**.
- **Socializing**: social media changed how we find partners, get news and organize for political change. Sites include Facebook, Twitter, LinkedIn, YouTube, Flickr, Second Life, Delicious, blogs and wikis. Facebook seems to be the leading place to share opinions.
- The **internet** is the international network of interconnected computer networks. Its main purpose is the quest for information (browsing, e-mail, newsgroups, file transfer).

KEY FACTS

Flipped classroom Lectures watched at home, exercises done in class	**Pharmacy robots** Pick drugs so staff can dispense and check medication
Job hunting Done from home with a phone, laptop or desktop	**Leading social site** Facebook (per the module)

Socialization How a child adapts behavior to be an effective member of society, through learning or education	

Explain like I'm 5: Everyday ICT is everywhere. You message friends, learn in class with screens, and even search for jobs on your phone. Social media is like a giant town square where everyone shares what they think.

3

MODULE 1: INTRODUCTION TO ICT

Impact of ICT in Society

- **Positive impacts**: better and cheaper communication (VoIP, instant messaging, e-mail, video conferencing), improved access to education (distance learning, online tutorials, interactive multimedia, virtual reality), and new tools such as digital cameras, photo-editing software and screen-reading software for the blind.
- More positives: information management (data mining for targeted advertising, better stock control), security (encryption keeps data safe while stored or sent), 'impossible' experiments through simulations, access to worldwide markets and society, and new jobs such as systems analysts, programmers, software engineers and help desk operators.
- **Negative impacts**: job loss, reduced personal interaction, reduced physical activity, cost, and competition.
- **Job loss** comes from automation (robots on an assembly line), job export (data processing sent to cheaper countries), and fewer workers doing the same work (a bar-code scanner at the checkout).
- **Reduced physical activity** can lead to obesity, heart disease and diabetes.

POSITIVES VS NEGATIVES

Positive: Education Distance learning, online tutorials, VR	**Positive: Communication** VoIP, e-mail and video conferencing save cost
Positive: Security Encryption protects stored and sent data	**Positive: Jobs** Systems analysts, programmers, help desk
Negative: Job loss Automation, job export, fewer workers needed	**Negative: Interaction** Less social contact can cause isolation
Negative: Health Less activity: obesity, heart disease, diabetes	**Negative: Cost** Expensive hardware, software and specialist staff
Negative: Competition Rivals may offer the same service for less	

Explain like I'm 5: ICT is like a superpower with side effects. It makes talking, learning and working easier and cheaper. But machines can take over some jobs, we may sit too much, and fancy gadgets cost a lot.

4 MODULE 2: HISTORY OF COMPUTERS
Definition and Characteristics of a Computer

- A **computer** is a programmable machine.
- It is an electronic device that manipulates information, or data. It can store, retrieve and process data.
- It manipulates data according to a list of instructions called a **program**.
- It is any device that aids humans in performing computations or calculations.
- The earliest 'computers' were humans whose job title was computer. They worked out mathematical expressions after years of math training. The word 'computer' was first recorded in **1613** for a person who did calculations, and kept that meaning until the middle of the 20th century.

THREE PRINCIPAL CHARACTERISTICS

Characteristic 1 Responds to a specific set of instructions in a well-defined manner	**Characteristic 2** Can execute a pre-recorded list of instructions
Characteristic 3 Can quickly store and retrieve large amounts of data	

Explain like I'm 5: A computer is a super fast helper that follows instructions exactly. It cannot think by itself. You give it a list of steps, and it remembers a lot and works very quickly. Long ago, 'computer' was even the job name of people who did math all day.

5 MODULE 2: HISTORY OF COMPUTERS
Applications of ICT (Computers) in Daily Life

- The module lists 20 areas where computers are used.
- Business: store accounts and records, manage projects, track inventory, make presentations and reports, and talk with customers.
- Education: audio-visual packages, interactive exercises, e-books, online exams and tutoring over the internet.
- Navigation: computers combined with GPS and satellites pinpoint your exact location.
- Robotics: one of the first uses was in manufacturing to build cars.

THE 20 APPLICATIONS

Business Accounts, records, projects, inventory, e-mail	**Education** E-learning, e-books, online exams, tutoring
Healthcare Digital patient data, diagnosis aid, lab and heart monitors	**Retail and Trade** Buy and sell online, compare prices (eBay, Craigslist)
Government City planning, law enforcement, traffic, tourism	**Marketing** Data-driven campaigns, websites, social media
Science Research, data analysis, controlling spacecraft	**Publishing** Newsletters, magazines, novels, e-books
Arts and Entertainment Drawing, photos, music, video, games	**Communication** Video calls, e-mail, videoconferencing (Skype)
Banking and Finance Check balances, transfer money, trade stocks	**Transport** Safety and navigation systems, automated driving
Navigation Computers plus GPS and satellites show your location	**Working From Home** Remote work and remote monitoring

Military Training, intelligence, smart weapons, drones	**Social and Romance** Social media, dating apps, blogs, forums
Booking Vacations Timetables, tickets, hotels, Air BnB	**Security and Surveillance** Biometric passports, face recognition, speed cameras
Weather Forecasting Processes huge amounts of meteorological data	**Robotics** Machines for harsh or specific jobs, first used to build cars

Explain like I'm 5: Computers are helpers in almost every job: banks, hospitals, schools, shops, planes, phones, and even weather reports. Anywhere there is lots of information to store, send or calculate, there is probably a computer.

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MODULE 2: HISTORY OF COMPUTERS

Early Computing Devices (Tally Sticks to Arithmometer)

- Calculations were first done by humans. Early devices were memory and calculation aids.
- Remember who invented what and when: Napier (1614), Oughtred (1622), Pascal (1642), Leibniz (1672), de Colmar (1820).
- The **Pascaline** was limited to addition and subtraction and was too expensive.
- The **Stepped Reckoner** could add, subtract, multiply and divide automatically.
- The **Arithmometer** was the first reliable, useful, commercially successful and mass-produced calculating machine.

TIMELINE

Ancient	**Tally sticks**: a memory aid to record numbers, quantities or even messages.
2400 B.C.	**Abacus**: invented in Babylonia. The familiar form was first used in China around 500 B.C. Does basic arithmetic.
1614	**Napier's Bones** (John Napier): multiply, divide and find square and cube roots by moving rods in special boards.
1622	**Slide Rule** (William Oughtred): based on Napier's logarithms. For multiplication, division, roots, logarithms and trigonometry, not normally addition or subtraction.
1642	**Pascaline** (Blaise Pascal): addition and subtraction only, and too expensive.
1672	**Stepped Reckoner** (Gottfried Wilhelm Leibniz): adds, subtracts, multiplies and divides automatically.
1820	**Arithmometer** (Thomas de Colmar): first reliable, commercially successful, mass-produced calculating machine. Does the four basic operations.
1881*	**Jacquard Loom** (Joseph-Marie Jacquard): an automatic loom controlled by punched cards. (*Year as written in your module.)

Explain like I'm 5: Before electricity, people built clever tools to count: sticks with notches, beads on rods, and gear machines. Each new tool did a bit more math than the last, until one machine could add, subtract, multiply and divide by itself.

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MODULE 2: HISTORY OF COMPUTERS

From Babbage to the First Computer Companies

- Memorize the 'firsts': first mechanical computer (Babbage), first programmer (Ada Byron), first printing calculator (Scheutz), first programmable computer (Z1), first electronic digital device (ABC), first electro-mechanical computer (Harvard Mark 1), first electronic general-purpose computer (ENIAC), first commercial computer (UNIVAC I), first stored program computer (EDVAC), first portable computer (Osborne 1).
- **Eckert and Mauchly** developed ENIAC, designed UNIVAC I and founded the first computer company.
- Acronyms: **ENIAC** = Electronic Numerical Integrator and Computer. **EDVAC** = Electronic Discrete Variable Automatic Computer. **UNIVAC** = UNIVersal Automatic Computer.

TIMELINE

1822 and 1834	**Difference Engine and Analytical Engine** (Charles Babbage): automatic mechanical calculator for polynomial functions. The first mechanical computer.
1840	**First computer programmer**: Augusta Ada Byron suggested the binary system to Babbage and wrote programs for the Analytical Engine.
1843	**Scheutzian Calculation Engine** (Per Georg Scheutz): based on Babbage's difference engine. The first printing calculator.
1890	**Tabulating Machine** (Herman Hollerith): helps summarize information and accounting.
1936 to 1938	**Z1** (Konrad Zuse, Germany): the first programmable computer. Used punch tape for input and output.
1939 to 1942	**Atanasoff-Berry Computer, ABC** (John Atanasoff and Clifford Berry, Iowa State University): the first electronic digital computing device.
1943	**Harvard Mark 1** (Howard H. Aiken), also called IBM ASCC: the first electro-mechanical computer.
1946	**ENIAC** (Eckert and Mauchly): the first electronic general-purpose computer.
1949	**Electronic Controls Company**: the first computer company, founded by Eckert and Mauchly.
Commercial era	**UNIVAC I** (Eckert and Mauchly): the first commercial computer.
1952	**EDVAC** (Von Neumann): the first stored program computer. Its memory holds both a stored program and data.
1981	**Osborne 1**: the first portable computer, released by the Osborne Computer Corporation.

Explain like I'm 5: Think of a relay race of inventors. Babbage drew the first computer plan, Ada wrote the first program for it, and later teams built electric ones. ENIAC was the big electronic star, UNIVAC was the first one sold to businesses, and Osborne 1 was the first you could carry.

8 MODULE 2: HISTORY OF COMPUTERS
Basic Computing Periods (Ages)

- There are four ages of information technology: **Premechanical**, **Mechanical**, **Electromechanical** and **Electronic**. The electronic age is the age we live in now.
- Premechanical: language and petroglyphs (picture drawings carved in rock), the Phoenician alphabet, marks in wet clay, papyrus paper, Chinese paper from rags, the first books and libraries, and Egyptian scrolls.
- The first numbering system (1 to 9) came from India around 100 A.D. The number **0** was invented in 875 A.D. The abacus was the popular calculator.
- Harvard Mark 1 size: 8 ft high, 50 ft long, 2 ft wide, 5 tons, programmed with punch cards.
- ENIAC: the first high-speed, digital, reprogrammable computer. Made for the U.S. Army for artillery firing tables. It covered 680 square feet, weighed 30 tons and used vacuum tubes.

THE FOUR AGES

3000 B.C. to 1450 A.D.	**Premechanical**: petroglyphs, alphabets, paper, books, libraries, numbers, abacus.
1450 to 1840	**Mechanical**: slide rule, Pascaline and Babbage's difference engine (tabulates polynomial equations using finite differences).
1840 to 1940	**Electromechanical**: telegraph (early 1800s), Morse code by Samuel Morse (1835), telephone by Alexander Graham Bell (1876), radio by Guglielmo Marconi (1894), Harvard Mark 1.
1940 to now	**Electronic**: ENIAC and everything after it, up to today's computers.

Explain like I'm 5: Computer history has four chapters. First people drew and counted by hand. Then they built gear machines. Then electricity brought the telegraph, telephone and radio. Now we live in the electronic chapter with fast digital computers.

9 MODULE 2: HISTORY OF COMPUTERS
Generations of Computers

- Five generations: 1st 1946 to 1958, 2nd 1959 to 1964, 3rd 1965 to 1970, 4th 1971 to today, 5th today to the future.
- Shortcut to remember the technology: **vacuum tube**, **transistor**, **integrated circuit**, **microprocessor**, **artificial intelligence**.
- First generation machines used machine language, solved one problem at a time, took input from punched cards and paper tape and gave printouts. They were huge, hot and expensive.
- One transistor replaced the equivalent of 40 vacuum tubes.

TIMELINE

1st: 1946 to 1958	**Vacuum tubes** and magnetic drums. Machine language. Examples: ENIAC, EDSAC, UNIVAC I, II, 1101.
2nd: 1959 to 1964	**Transistors**: smaller, faster, cheaper, more reliable. Assembly language, magnetic core memory, FORTRAN and COBOL. Examples: UNIVAC III, RCA 501, IBM 7030 Stretch.
3rd: 1965 to 1970	**Integrated circuits** on silicon chips (semiconductors). Instructions in billionths of a second. Operating system and BASIC appeared.
4th: 1971 to today	**Microprocessor**: thousands of integrated circuits on one chip (CPU). Networks led to the Internet. GUI, mouse, handheld devices and the personal computer (Apple II).

5th: today to future	**Artificial Intelligence**: parallel processing and superconductors. Goal: natural language input, learning and self-organization. Voice recognition is an example.

Explain like I'm 5: Each generation made computers smaller and smarter. First came huge glowing tubes, then tiny transistors, then chips, then one chip that holds a whole brain, and now we are teaching computers to learn.

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MODULE 3: THE WEB AND THE INTERNET

The Web: Web 1.0, 2.0 and 3.0

- The **World Wide Web** is information organized into web pages with text and graphics, connected by **hypertext** links.
- A **website** is a collection of linked pages with a common theme. The **home page** is the main page that the other pages link back to.
- **Sir Tim Berners-Lee** (TimBL) is best known as the inventor of the World Wide Web.
- **Web 1.0**: read-only, static web. Used as an 'information portal', with tables for layout. Disadvantages: read-only, limited user interaction, lack of standards. Examples: Mp3.com, home pages, directories, page views, HTML portals.
- **Web 2.0**: read-write, interactive web with user-generated content and collaboration. Pages that react to user input are **dynamic pages**.
- **Web 3.0**: read-write intelligent web, named by John Markoff of the New York Times. Uses semantic web, distributed databases, natural language processing, machine learning, machine reasoning and autonomous agents.
- The **semantic web** is a web of data that systems can read and categorize, not just humans.

WEB 2.0 EXAMPLES AND FEATURES

Social networking Facebook, Twitter, LinkedIn, Google+, Pinterest, Tumblr, Instagram	**Blogs** WordPress, Blogger, Tumblr. Newest post appears first
Wikis Wikipedia, Wikibooks, Wiktionary, Wikinews and others. Edited by the audience	**Video sharing** YouTube, Facebook, LinkedIn, Flickr, Photobucket
Folksonomy Users classify information with freely chosen keywords (tagging)	**Rich user interface** Dynamic content that responds to user input
User participation Others add content through comments, reviews and evaluations	**Long tail** Services offered on demand, like a data plan

Explain like I'm 5: Web 1.0 is a library where you can only read books. Web 2.0 lets you write in the books too, like comments and posts. Web 3.0 is a smart library where the books understand what you want and bring you the right one.

11 MODULE 3: THE WEB AND THE INTERNET
Types of Websites

- The module lists 12 types of websites. Match each type with its purpose.

THE 12 TYPES

eCommerce Sells products directly. Has a shopping cart and payment	**Business** Represents a specific business, branded like it
Entertainment Visited purely for fun	**Portfolio** Shows examples of past work
Media Collects news stories and reporting	**Brochure** Simple business site with a few pages and contact info
Nonprofit Online presence and donations for a nonprofit	**Educational** Educational materials or information about an institution
Infopreneur Creates and sells information products: courses, tutorials, videos, eBooks	**Personal** Personal blogs, vlogs and photo diaries
Web portal Internal site gathering information from many sources, usually with login	**Wiki or forum** Many users collaborate on content, like Wikipedia

Explain like I'm 5: Websites are like different kinds of shops and places. Some sell things, some show your work, some tell news, some teach, and some are just your diary. The type depends on what the site is for.

12 MODULE 3: THE WEB AND THE INTERNET
The Internet: History and Major Components

- The **Internet** ('net', a network of networks) is the largest computer network in the world and connects billions of users. The word comes from 'interconnection' and 'network'. Generally, nobody owns the internet.
- A **network** is a collection of computers and devices connected by communication channels and transmission media to share resources.
- Major components: **servers**, **IP address**, **browser**, **DNS** and **ISP**.
- Server types: application, web, proxy, mail, file and policy.
- **Vinton Cerf** is one of the 'fathers of the Internet', sharing the title with TCP/IP co-developer Bob Kahn.

HISTORY TIMELINE

Jan 2, 1969	**ARPA** (Advanced Research Project Agency) starts an experimental computer network. No central server, equal importance for every computer, so losing one or two nodes does not break the network.
1982	The word 'internet' starts to be used.
1986	First 'free net' created at Case Western Reserve University.
1991	The US government allows business agencies to connect to the internet.

MAJOR COMPONENTS

Application server Provides the business logic for an application program	**Web server** Serves requested HTML pages or files
Proxy server Intermediary between an endpoint device and another server	**Mail server** Receives incoming e-mail and forwards outgoing e-mail

File server Central storage and management of data files	**Policy server** Security component that provides authorization services and tracks and controls files
IP address Numerical label that identifies each network device	**Browser** Program for viewing information on the web: Chrome, Safari, Internet Explorer, Opera, Mozilla
DNS Phonebook of the internet. Examples: www.facebook.com, www.pup.edu.ph, www.academia.edu	**ISP** Provides access to the internet. Two types: national and regional. Examples: Sky Broadband, PLDT, Converge

TOP-LEVEL DOMAINS

.com Commercial	**.org** Organization
.net Network	**.edu** Education
.gov National and state government agencies	**.ph** Philippines
.au Australia	

Explain like I'm 5: The internet is a giant web of connected computers that nobody owns. Your device has an address (IP), a phonebook (DNS) turns names like facebook.com into addresses, a browser opens pages, and your ISP is the company that connects you.

MODULE 3: THE WEB AND THE INTERNET

13 Uses of the Internet and Internet Terms

- Uses: look for information; school, jobs and home; e-mail; video teleconferencing; buy and sell products; social networking; watch and post videos; games; take college courses; monitor home while away; financial transactions; download music and movies.
- A **bit** is a single binary digit (0 or 1). A **byte** is generally eight bits.
- **HTTP** is the data communication standard of web pages. **HTTPS** adds a layer of encryption.
- **Upload** = send data from your computer to another computer. **Download** = bring data from another computer to yours.

TERMS AND DEFINITIONS

Internet Global network of thousands of computer networks linked by data lines and wireless systems	**Web** Collection of billions of webpages viewed with a web browser
Email Most common method of sending and receiving messages online	**Social media** Sites and apps for sharing comments, photos and videos
Online gaming Games played with and against others over the Internet	**Software updates** OS and app updates typically downloaded from the Internet
HTML Hypertext Markup Language: coding language that tells a browser how to place pictures, text, multimedia and links	**URL** Uniform Resource Locator: web address of a remote resource
Bit Single digit in the binary system (base 2): 0 or 1	**Byte** Generally eight bits

Upload Transfer data from your computer to another computer	**Download** Transfer data from another computer to your computer
HTTP Hypertext Transfer Protocol: data communication standard of web pages	**HTTPS** HTTP Secure: adds encryption to hide personal information and passwords
Router Acts as the traffic cop for network signals coming from your ISP. Wired, wireless or both	**Encryption** Mathematical scrambling of data so eavesdroppers cannot read it
Web bot Program that performs a repetitive task on the Internet, like searching for information	**Search engine** Software like Google or Yahoo that searches the web using keywords and phrases

Explain like I'm 5: These are the everyday words of the internet. A bit is one tiny 0 or 1, and eight of them make a byte. Uploading sends things out from you, downloading brings things to you, and the little 's' in https means your messages are put in a secret code.

Practice Exam

40 items. Circle the letter of the correct answer. The answer key is at the end.

1. ICT stands for:

- **A.** Information and Communications Technology
- **B.** Internet and Computer Technology
- **C.** Information and Computing Tools
- **D.** Integrated Communication Terminals

2. Which best describes ICT?

- **A.** Only computer hardware
- **B.** All devices, networks, applications and systems that let people and organizations interact in the digital world
- **C.** Only the internet and social media
- **D.** Only mobile phones

3. In the 'flipped classroom' model, students:

- **A.** Only use chalkboards and textbooks
- **B.** Do all their work in class without computers
- **C.** Watch lectures at home on the computer and use class time for interactive exercises
- **D.** Are taught only by robots

4. Why do many pharmacies use robot technology to pick prescribed drugs?

- **A.** To make medicines free
- **B.** To avoid using any computers
- **C.** To replace all pharmacists
- **D.** So highly trained staff can focus on jobs needing human intelligence and interaction

5. According to the module, which social site seems to be the leading place where people communicate and share opinions?

- **A.** Facebook
- **B.** Flickr
- **C.** LinkedIn
- **D.** Delicious

6. Which is a POSITIVE impact of ICT?

- **A.** Job loss through automation
- **B.** High cost of hardware and software
- **C.** Improved access to education through distance learning and online tutorials
- **D.** Reduced personal interaction

7. Robots replacing people on an assembly line is an example of:

- **A.** Job loss (a negative impact)
- **B.** Creation of new jobs
- **C.** Better security through encryption
- **D.** Distance learning

8. Reduced physical activity caused by heavy ICT use can lead to:

- **A.** Obesity, heart disease and diabetes
- **B.** Better social skills
- **C.** Cheaper hardware
- **D.** Faster data processing

9. Which is a principal characteristic of a computer?

- **A.** It can think and feel like a human
- **B.** It only works when connected to the internet
- **C.** It can quickly store and retrieve large amounts of data
- **D.** It works without any instructions

10. The first recorded use of the word 'computer' (1613) referred to:

- **A.** A mechanical calculator
- **B.** A punch card machine
- **C.** An electronic device
- **D.** A person who carried out calculations

11. Computers that control heart rate monitors and help analyze data to find diagnoses belong to which application?

A. Marketing
B. Healthcare
C. Publishing
D. Navigation

12. Computers combined with satellites and GPS to pinpoint your location are used in:

A. Publishing
B. Retail and trade
C. Marketing
D. Navigation

13. Who invented the slide rule in 1622, based on Napier's ideas about logarithms?

A. William Oughtred
B. Blaise Pascal
C. Thomas de Colmar
D. Gottfried Leibniz

14. The Pascaline (1642), limited to addition and subtraction, was invented by:

A. Herman Hollerith
B. Charles Babbage
C. John Napier
D. Blaise Pascal

15. Which Leibniz machine (1672) could add, subtract, multiply and divide automatically?

A. Stepped Reckoner
B. Abacus
C. Napier's Bones
D. Pascaline

16. A byte generally consists of:

A. Sixteen bits
B. Eight bits
C. Two bits
D. Four bits

17. Charles Babbage's Difference Engine and Analytical Engine are considered the first:

A. Mechanical computer
B. Commercial computer
C. Portable computer
D. Electronic computer

18. Who wrote programs for the Analytical Engine and is called the first computer programmer?

A. Per Georg Scheutz
B. Augusta Ada Byron
C. Clifford Berry
D. Konrad Zuse

19. Which was the first electro-mechanical computer (also known as the IBM ASCC)?

A. UNIVAC I
B. Harvard Mark 1
C. ENIAC
D. Z1

20. The first programmable computer, created by Konrad Zuse in Germany, was the:

A. Osborne 1
B. Z1
C. EDVAC
D. ABC

21. The first electronic digital computing device, built at Iowa State University, was the:

A. ENIAC
B. Atanasoff-Berry Computer (ABC)
C. Harvard Mark 1
D. UNIVAC I

22. ENIAC stands for:

A. Electronic Number Input and Computer
B. Electronic Numerical Integrator and Calculator
C. Electronic Numerical Integrator Automatic Computer
D. Electronic Numerical Integrator and Computer

23. The first commercial computer was the:

A. Harvard Mark 1
B. Z1
C. ENIAC
D. UNIVAC I

24. The first stored program computer, designed by Von Neumann, was the:

A. ENIAC
B. EDVAC
C. Z1
D. UNIVAC I

25. Which age ran from about 3000 B.C. to 1450 A.D. and included petroglyphs and the abacus?

A. Mechanical
B. Electronic
C. Premechanical
D. Electromechanical

26. The telegraph, Morse code, telephone and radio belong to which age (1840 to 1940)?

A. Electronic
B. Premechanical
C. Mechanical
D. Electromechanical

27. First generation computers (1946 to 1958) used:

A. Vacuum tubes
B. Microprocessors
C. Transistors
D. Integrated circuits

28. In the second generation, one transistor replaced the equivalent of about how many vacuum tubes?

A. 400
B. 40
C. 4
D. 1,000

29. The integrated circuit was the hallmark of which generation?

A. First generation
B. Fourth generation
C. Third generation
D. Second generation

30. Fifth generation computers are based on:

A. Artificial Intelligence
B. Magnetic drums
C. Punched cards
D. Vacuum tubes

31. Web 1.0 is best described as the:

A. Read-write interactive web
B. Read-write intelligent web
C. Read-only, static web
D. Mobile-only web

32. Who is best known as the inventor of the World Wide Web?

A. Bob Kahn
B. Sir Tim Berners-Lee
C. Vinton Cerf
D. John Markoff

33. Web pages that are affected by user input or preference, like comments or accounts, are called:

A. Dynamic pages
B. Home pages
C. Static pages
D. Semantic pages

34. The semantic web is described as a:

A. Web of static pages
B. Web of home pages only
C. Web of data
D. Web of printed documents

35. A website with a shopping cart and a way to pay by credit card is a(n):

A. eCommerce website
B. Brochure website
C. Portfolio website
D. Personal website

36. Websites devoted to showing examples of past work are called:

A. Nonprofit websites
B. Media websites
C. Portfolio websites
D. Web portals

37. ARPA started an experimental computer network on January 2 of which year?

A. 1991
B. 1982
C. 1969
D. 1986

38. The Domain Name System (DNS) is often called the:

A. Phonebook of the internet
B. Library of the web
C. Post office of the internet
D. Memory of the browser

39. Which domain suffix stands for the Philippines?

A. .edu
B. .au
C. .gov
D. .ph

40. HTTPS differs from HTTP because it:

A. Adds a layer of encryption to hide your personal information
B. Only works on mobile phones
C. Blocks all advertisements
D. Makes pages load faster

Answer Key

#	Answer	Review topic
1	**A** Information and Communications Technology	Topic 1
2	**B** All devices, networks, applications and systems that let people and organizations interact in the digital world	Topic 1
3	**C** Watch lectures at home on the computer and use class time for interactive exercises	Topic 2
4	**D** So highly trained staff can focus on jobs needing human intelligence and interaction	Topic 2
5	**A** Facebook	Topic 2
6	**C** Improved access to education through distance learning and online tutorials	Topic 3
7	**A** Job loss (a negative impact)	Topic 3
8	**A** Obesity, heart disease and diabetes	Topic 3
9	**C** It can quickly store and retrieve large amounts of data	Topic 4
10	**D** A person who carried out calculations	Topic 4
11	**B** Healthcare	Topic 5
12	**D** Navigation	Topic 5
13	**A** William Oughtred	Topic 6
14	**D** Blaise Pascal	Topic 6
15	**A** Stepped Reckoner	Topic 6
16	**B** Eight bits	Topic 13
17	**A** Mechanical computer	Topic 7
18	**B** Augusta Ada Byron	Topic 7
19	**B** Harvard Mark 1	Topic 7
20	**B** Z1	Topic 7
21	**B** Atanasoff-Berry Computer (ABC)	Topic 7
22	**D** Electronic Numerical Integrator and Computer	Topic 7
23	**D** UNIVAC I	Topic 7
24	**B** EDVAC	Topic 7
25	**C** Premechanical	Topic 8
26	**D** Electromechanical	Topic 8
27	**A** Vacuum tubes	Topic 9
28	**B** 40	Topic 9
29	**C** Third generation	Topic 9
30	**A** Artificial Intelligence	Topic 9
31	**C** Read-only, static web	Topic 10
32	**B** Sir Tim Berners-Lee	Topic 10
33	**A** Dynamic pages	Topic 10
34	**C** Web of data	Topic 10
35	**A** eCommerce website	Topic 11
36	**C** Portfolio websites	Topic 11
37	**C** 1969	Topic 12
38	**A** Phonebook of the internet	Topic 12
39	**D** .ph	Topic 12
40	**A** Adds a layer of encryption to hide your personal information	Topic 13